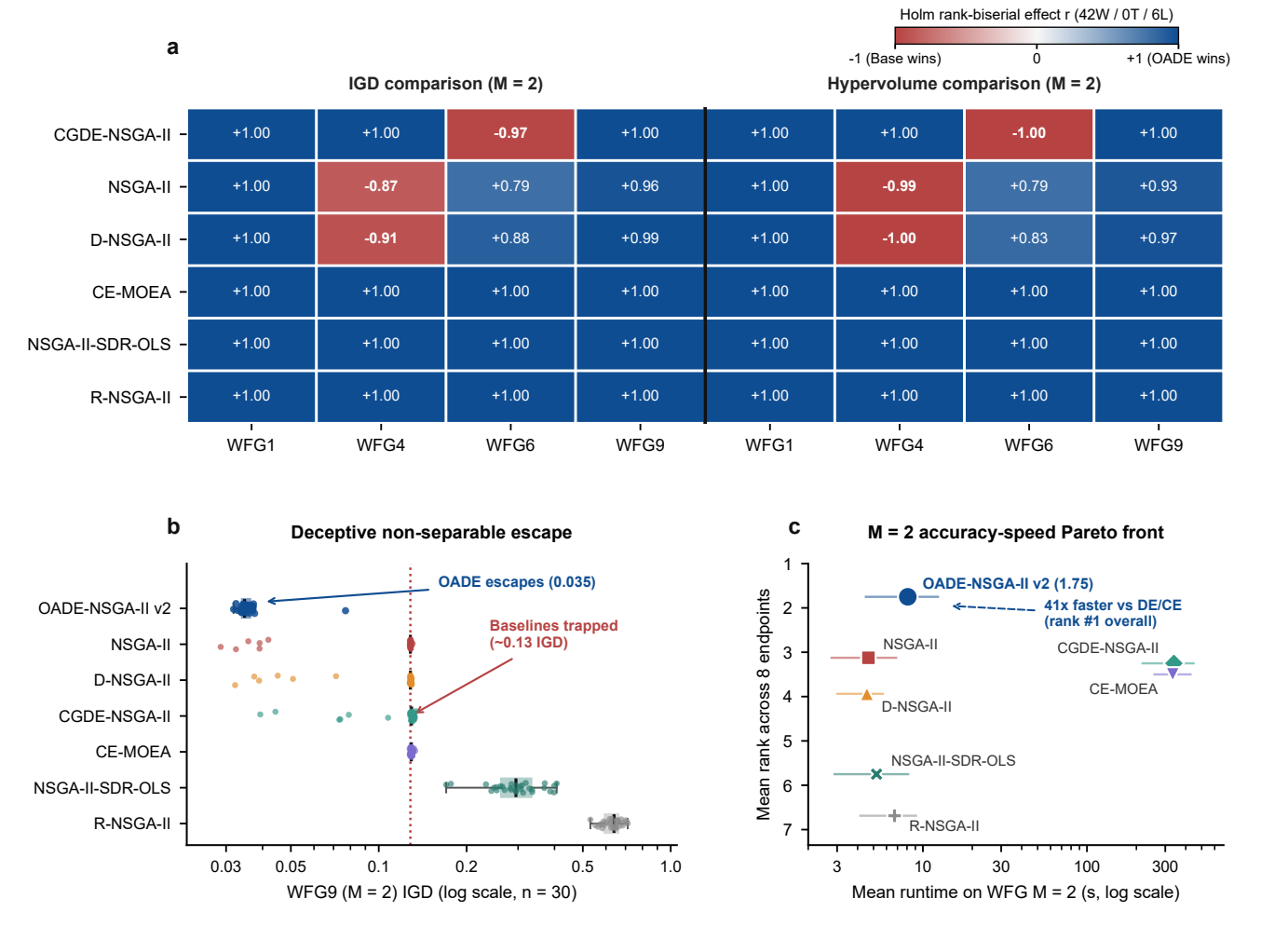

a
Holm rank-biserial effect r (42W / 0T / 6L)
-1 (Base wins)
0
+1 (OADE wins)
IGD comparison (M = 2)
Hypervolume comparison (M = 2)
CGDE-NSGA-II
NSGA-II
D-NSGA-II
CE-MOEA
NSGA-II-SDR-OLS
R-NSGA-II
+1.00
-0.97
-0.87
+0.79
+0.96
-0.91
+0.88
+0.99
-1.00
-0.99
+0.83
+0.97
+0.93
WFG1
WFG4
WFG6
WFG9
b
Deceptive non-separable escape
OADE escapes (0.035)
Baselines trapped
(~0.13 IGD)
OADE-NSGA-II v2
NSGA-II
D-NSGA-II
CGDE-NSGA-II
CE-MOEA
NSGA-II-SDR-OLS
R-NSGA-II
0.03
0.05
0.1
0.2
0.5
1.0
WFG9 (M = 2) IGD (log scale, n = 30)
c
M = 2 accuracy-speed Pareto front
OADE-NSGA-II v2 (1.75)
41x faster vs DE/CE
(rank #1 overall)
NSGA-II
CGDE-NSGA-II
CE-MOEA
D-NSGA-II
NSGA-II-SDR-OLS
R-NSGA-II
Mean rank across 8 endpoints
1
2
3
4
5
6
7
3
10
30
100
300
Mean runtime on WFG M = 2 (s, log scale)